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TetraTHON 2026

An Indo-French AI Innovation Sprint · Navrachana Innovation Foundation (NIF)

PROJECT STATEMENT OF WORK

TEAM & SUBMISSION DETAILS

Team Name	360 HORIZON (TETRA004)
Team Lead Name	MAPTUE FOTSO Lyne Vanelle
Team Lead Email	lyne-vanelle.maptue-fotso@isen.yncrea.fr
Team Lead Phone	+33618604954
Date Submitted	02/08/2026

SECTOR TRACK

☒ HealthTech	☐ FinTech	☐ AgriTech	☐ EdTech
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PROJECT INFORMATION

Project / Solution Name	AI Risk Prediction & Early Triage Assistant (HT1)
Problem Statement No.	HT1 (HealthTech Track 1)
Industry Partner (if applicable)	ISEN Méditerranée / Navrachana University

ASSUMPTIONS

List any assumptions made about data availability, tools, APIs, or constraints.

Data Availability & Privacy: Benchmark clinical datasets (Pima Indians Diabetes, Framingham Heart Study, Stroke Prediction Dataset, UCI CKD) are sufficient for training multi-pathology screening models during the hackathon sprint. Access to credentialed repositories like PhysioNet or MIMIC-IV requires post-hackathon authorization. **Feature Spaces:** Clinical parameters vary across pathologies; hence, deploying four specialized machine learning models is more effective than a single multi-label model to avoid signal dilution. **Primary Care Deployment:** Primary care settings (e.g., in low-resource environments like India) rely on routine, non-invasive clinical indicators (demographics, vital signs, basic lab tests) rather than immediate advanced diagnostic tests. **Model Autonomy:** The platform acts strictly as a Clinical Decision Support System (CDSS) for screening and early triage; final medical diagnoses remain under the authority of clinical practitioners.



SCOPE OF WORK

Describe what the team will build during the 36-hour sprint, including key features and boundaries.

During the 36-hours sprint, Team 360 HORIZON built and deployed a functional clinical decision-support platform featuring: Clinical Data Entry Interface: Dynamic form for inputting patient demographics, vital signs, clinical history, and basic lab tests. Multi-Pathology Risk Scoring Engine: 4 specialized XGBoost classifiers predicting risk levels for Diabetes, Coronary Artery Disease (CAD), Stroke, and Chronic Kidney Disease (CKD). Consolidated Clinical Dashboard: Real-time risk stratification (Low / Moderate / High / Urgent) with automated visual alerts for critical thresholds. Explainable AI (XAI) Integration: Patient-level SHAP (SHapley Additive exPlanations) attribution graphs explaining the key risk drivers behind each score. Actionable Recommendations: Automated identification of missing diagnostic tests required to refine prediction accuracy. Out-of-Scope (Sprint Version): User authentication/RBAC, persistent Electronic Health Record (EHR) database storage, direct HIS/EHR live integration, and automated definitive diagnoses.

PROJECT GOALS

Minimize diagnostic delays for lifestyle-related chronic diseases (Diabetes, CAD, Stroke, CKD) in resource-constrained environments. Empower primary healthcare workers with interpretable, transparent AI risk assessment based on accessible routine clinical parameters. Achieve high recall/sensitivity for critical conditions (e.g., 82% recall on Stroke, 96% on early CKD screening) to ensure high-risk cases are caught early.

IMPACT & SCALABILITY

How does this solution support inclusive growth and scale beyond the sprint? (theme: AI for Inclusive Growth and Scalable Impact)

- Inclusive Healthcare Access: Designed specifically for low-resource primary care clinics where consultation time and specialized diagnostic tools are limited. By using basic non-invasive indicators, the platform enables early screening without requiring immediate expensive laboratory infrastructure.
- Scalable Architecture: Built on lightweight, decoupled microservices (FastAPI REST backend + React frontend) that can be deployed via web browsers or mobile devices in rural clinics.
- Modular Expansion: The core XGBoost + SHAP pipeline can easily be expanded to include additional cardiometabolic, respiratory, or tropical health conditions without restructuring the platform.

PROJECT DELIVERABLES

e.g. working prototype, demo video, pitch deck, source code repository

Working Prototype: Fully integrated frontend (React/Tailwind) dashboard and FastAPI backend pipeline.

Public Source Code Repository: github.com/boudjellal-malak/TETRA004



Technical Specification Document: System Requirements Specification (Cahier des Charges).
Presentation Deck: Pitch deck detailing clinical problem, ML methodology, and roadmap

TEAM STRUCTURE

Name	Role on Project	Institution	Email
Maptue Fotso Lyne Vanelle	Team Lead / Lead Developer	ISEN Méditerranée	lyne.maptue-fotso@isen-ouest.yncrea.fr
Malak Boudjellal	Machine Learning Engineer	ISEN Méditerranée	malak.boudjellal@isen.yncrea.fr
AMA Vivi EKOUVO	Full-Stack Developer	ISEN Méditerranée	ama-vivi.ekouvo@isen.yncrea.fr

PROPOSED TECH STACK

- Frontend: React (Vite), Tailwind CSS (Dynamic forms, SHAP visualization, risk dashboard).
- Backend API: FastAPI (Python 3.10+), RESTful API endpoints, Pydantic validation.
- Machine Learning Engine: XGBoost, Scikit-Learn (4 specialised disease models + 1 early CKD variant).
- Explainability (XAI): SHAP (TreeExplainer) for real-time feature attribution.

FUTURE COST TO SCALE

If this project were taken beyond the hackathon into a prototype or startup, rough order-of-magnitude cost to build and launch (not required to be precise).

Estimated Budget: \$25,000 – \$40,000 USD

EHR & HIPAA Compliance / Cloud Infrastructure (AWS/GCP Health): ~\$8,000 / year

Clinical Datasets Access & PhysioNet/MIMIC-IV Integration: ~\$5,000

Clinical Validation Study & Usability Review: ~\$10,000

Security & Authentication (OAuth2, Encryption, Testing): ~\$7,000

ACKNOWLEDGEMENT

By submitting this Statement of Work, the team confirms the information above is accurate and agrees to the TetraTHON 2026 judging criteria and event guidelines.

Team Lead Signature

Lyne Vanelle MAPTUE FOTSO



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Date

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